Name: Afifa Syed
Batch: T23
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EXPERIMENT NO. 07

Aim: To implement monitoring using Prometheus / Grafana / Nagios

LO Mapping: LO5

LO5: Implement ML lifecycle management and automated workflows using MLflow, Airflow, and monitoring tools like Prometheus/Grafana/Nagios.

Theory:

Continuous monitoring is a process to detect, report, respond all the attacks which occur in its infrastructure. Once the application is deployed into the server, the role of continuous monitoring comes in to play. The entire process is all about taking care of the company's infrastructure and respond appropriately.

Why We Need Nagios tool?

Here, are the important reasons to use Nagios monitoring tool:

- Detects all types of network or server issues
- Helps you to find the root cause of the problem which allows you to get the permanent solution to the problem
- Active monitoring of your entire infrastructure and business processes
- Allows you to monitor and troubleshoot server performance issues
- Helps you to plan for infrastructure upgrades before outdated systems create failures
- You can maintain the security and availability of the service
- Automatically fix problems in a panic situation
- Nagios is the most popular, open source, powerful monitoring system for any kind of infrastructure. It enables organizations to identify and resolve IT infrastructure problems before they affect critical business processes. Nagios has the capability of monitoring applications, services and entire IT infrastructure.
- NRPE is known as Nagios Remote Plugin Executor. It is designed to execute plugins on remote systems. On Windows hosts, the equivalent remote agent is NSClient++, which listens for Nagios requests and executes local checks (disk, CPU, memory, services, etc.).

Name: Afifa Syed
Batch: T23
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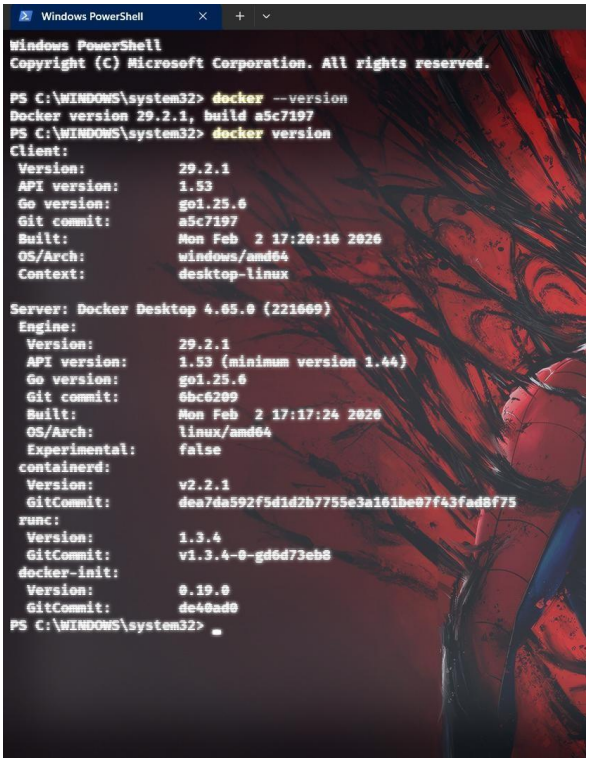
- NSClient++ runs as a Windows service on the remote host and waits for requests from the Nagios server (usually over the NRPE protocol, which NSClient++ also supports). When the Nagios server needs to check a resource on that host, it sends a command; NSClient++ executes the check locally and returns the result to Nagios.

Note on running Nagios on Windows

Nagios Core is built for Linux and does not install natively on Windows. The standard practice for a Windows machine is to run the Nagios Core server inside a Docker container (using Docker Desktop with the WSL2 backend), and to monitor Windows hosts using NSClient++, which is the Windows counterpart of NRPE. All commands below are Windows PowerShell / Command Prompt commands.

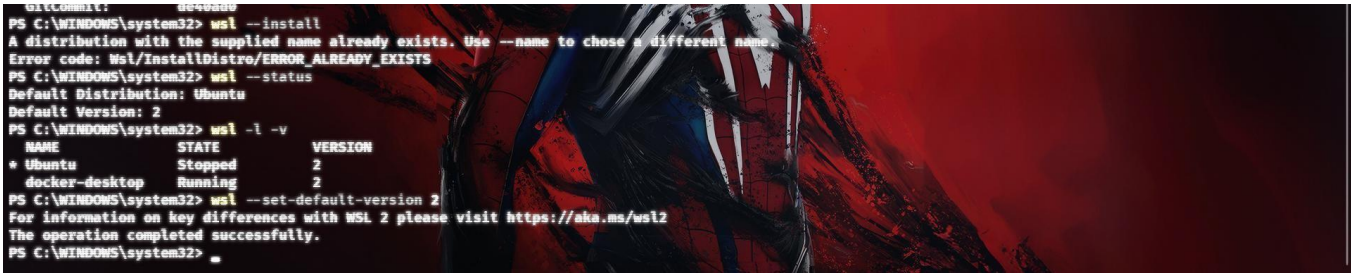
1 - Pre-requisite: Install Docker Desktop

Download and install Docker Desktop for Windows and enable the WSL2 based engine. Verify the installation from PowerShell:



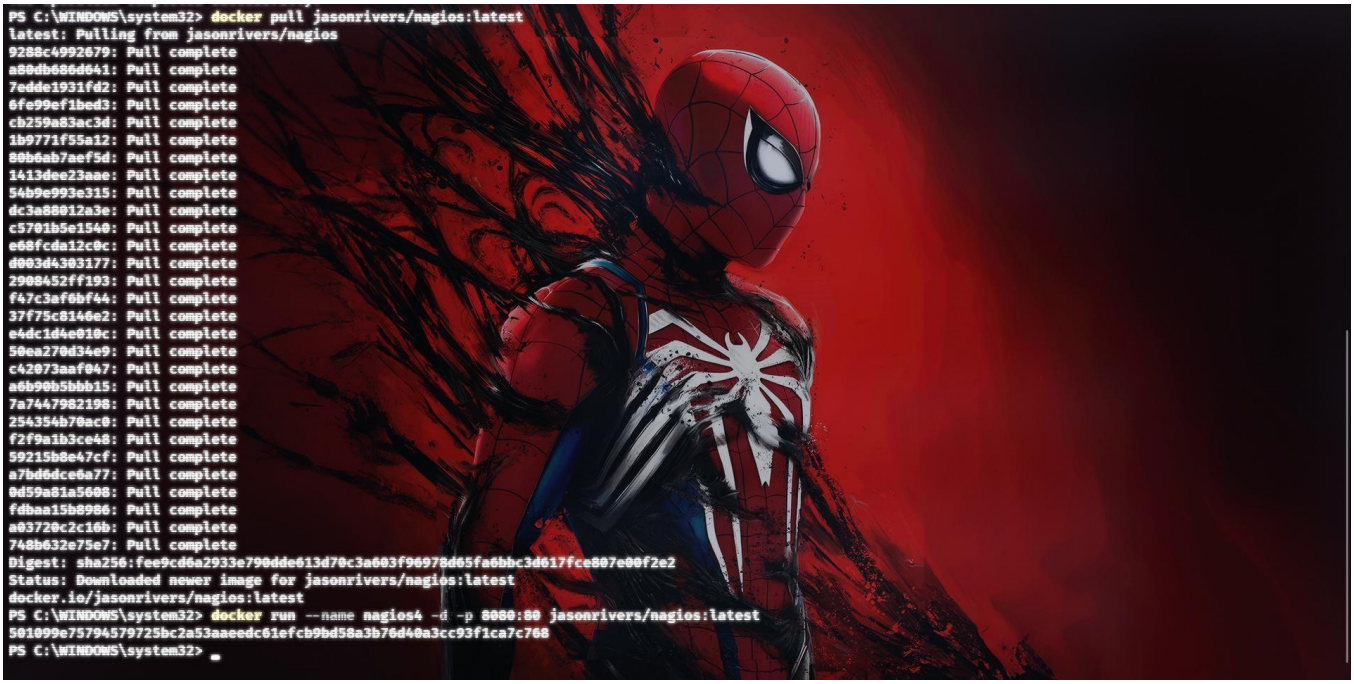
Also make sure the WSL2 feature is enabled:

Name: Afifa Syed
Batch: T23
Roll no :2511139



2 - Pull and Run the Nagios Core Docker Image

Pull a ready-made Nagios Core image and run it as a container, mapping the web port to the host.

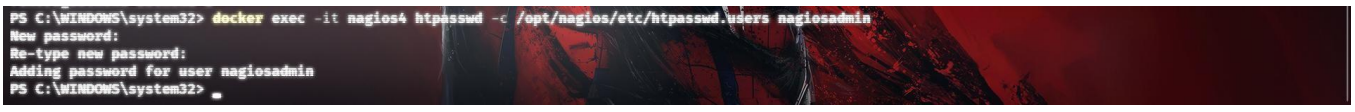


Verify the container is up and running:



3 - Set the Apache/Nagios Admin Password

Set (or reset) the password for the nagiosadmin user inside the running container:



Enter and confirm the password when prompted (e.g. 123456).

Name: Afifa Syed
Batch: T23
Roll no :2511139

4 - Access Nagios Web Interface

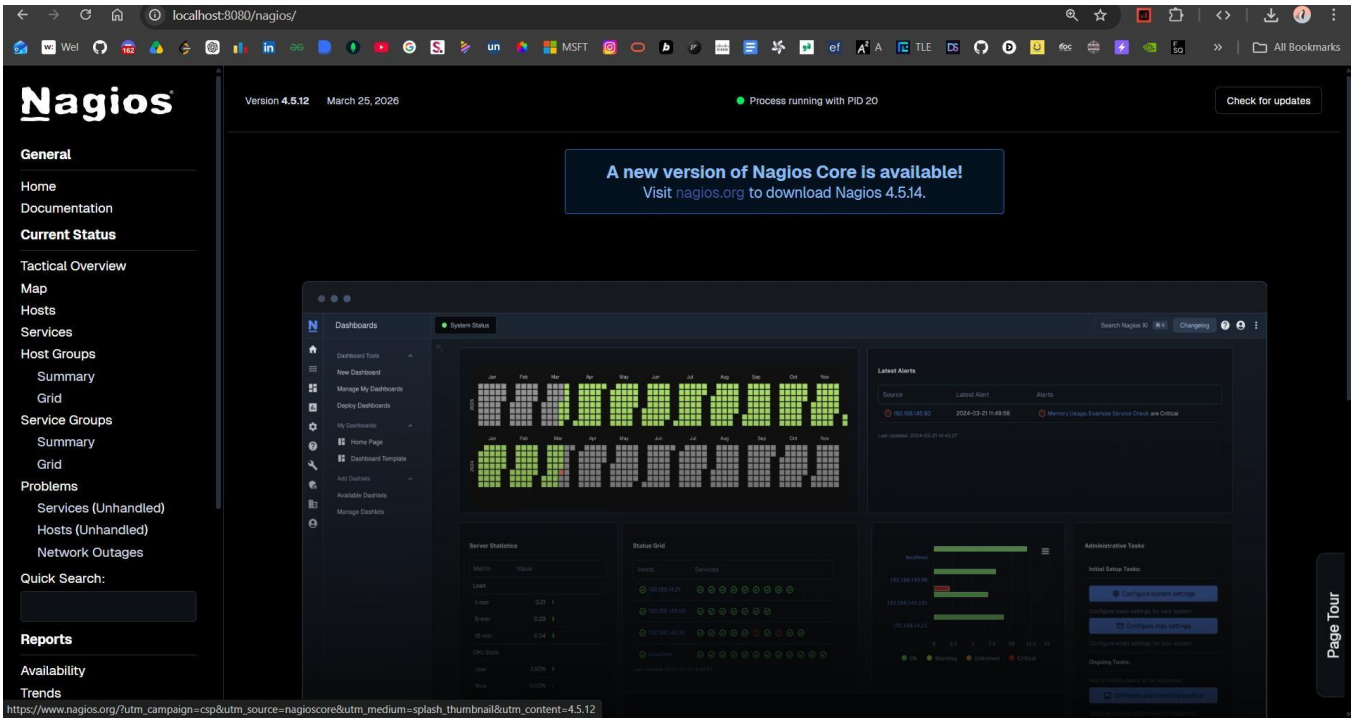
Access the Nagios web interface from a browser on the Windows host:

```
http://localhost:8080/nagios/
```

Prompting for Authentication Password - **username:**

nagiosadmin

Password: 123456 (which you entered while configuration)



5 - Install NSClient++ on the Windows Remote Host

On the Windows machine you want to monitor, download and silently install NSClient++ (the Windows equivalent of the NRPE agent):




```
PS C:\Users\vishal> Invoke-WebRequest -Uri
"https://github.com/mickem/nscp/releases/latest/download/NSCP-0.5.2-x64.msi"
OutFile "NSCP.msi"
PS C:\Users\vishal> Start-Process msiexec.exe -ArgumentList '/i NSCP.msi
```

Name: Afifa Syed
Batch: T23
Roll no :2511139

```
ADDLOCAL=ALL MODULES=CheckExternalScripts,NRPEServer /qn' -Wait
```

6 - Configure NSClient++

Edit the NSClient++ configuration file and add the Nagios server's IP address to the allowed hosts list:

Name: Afifa Syed
Batch: T23
Roll no :2511139

File Edit View

```
# If you want to fill this file with all available options run the following command:
#   nscp settings --generate --add-defaults --load-all
# If you want to activate a module and bring in all its options use:
#   nscp settings --activate-module <MODULE NAME> --add-defaults
# For details run: nscp settings --help


; in flight - TODO
[/modules]

; Load on startup - If the module should be loaded on startup
NRPEServer = enabled

; Load on startup - If the module should be loaded on startup
WEBServer = enabled

; Load on startup - If the module should be loaded on startup
CheckDisk = enabled

; Load on startup - If the module should be loaded on startup
CheckSystem = enabled

; Load on startup - If the module should be loaded on startup
CheckEventLog = enabled

; Load on startup - If the module should be loaded on startup
CheckHelpers = enabled

; Load on startup - If the module should be loaded on startup
CheckExternalScripts = enabled

; Load on startup - If the module should be loaded on startup
CheckNSCP = enabled


; in flight - TODO
[/settings/NRPE/server]

verify mode = none

tls version = tlsv1.2+

insecure = true

allowed ciphers = ADH


; in flight - TODO
[/settings/default]

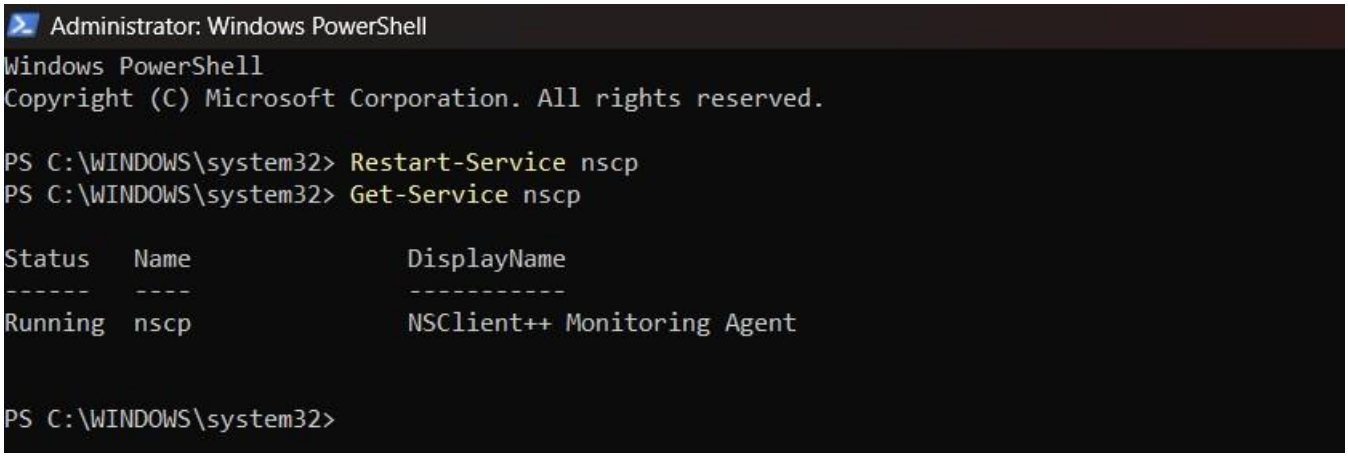
allowed hosts = 127.0.0.1,172.17.0.2
```

Ln 31, Col 1 | 1,269 characters | Plain text

Where 192.168.1.100 is your Nagios server (Docker host) IP address.

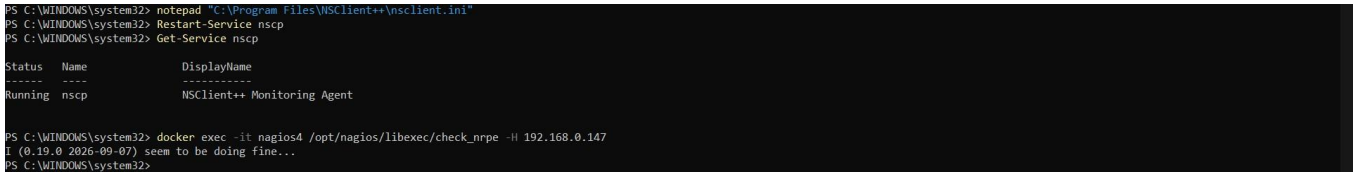
Name: Afifa Syed
Batch: T23
Roll no :2511139

After saving the file, restart the NSClient++ service so the change takes effect:



7 - Verify Connectivity from Nagios

From the Nagios server (inside the Docker container), run the NRPE check against the Windows remote host to confirm connectivity. Here 192.168.1.50 is the remote Windows host's IP address.



8 - Add the Windows Host in Nagios

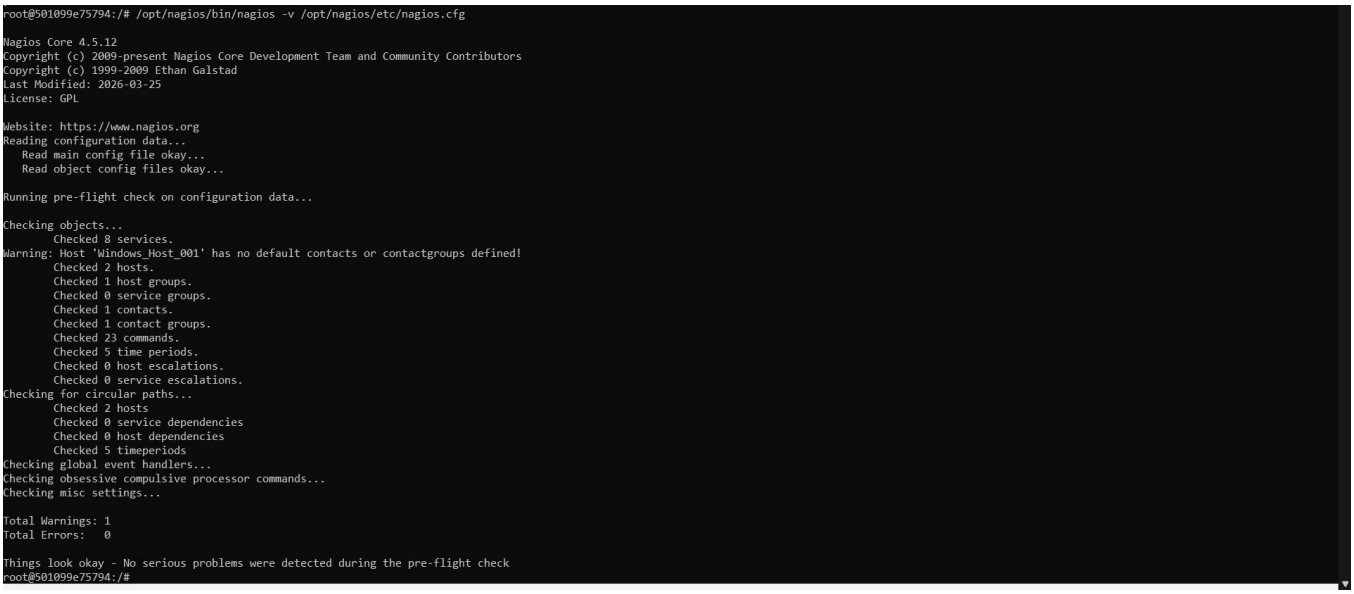
Create a configuration file for the new host inside the Nagios container. We also define a service that continuously checks whether the host is reachable (PING check).



Name: Afifa Syed
Batch: T23
Roll no :2511139

```
define host {
   use             windows-server     host_name
Windows_Host_001
alias           Windows Host 001     address
192.168.1.50
   register        1
}
define service{     host_name             Windows_Host_001     service_description
PING     check_command
check_ping!100.0,20%!500.0,60%     max_check_attempts
2     check_interval        2     retry_interval        2     check_period
24x7     check_freshness       1     contact_groups        admins
notification_interval  2     notification_period    24x7
notifications_enabled  1
   register             1
}
```

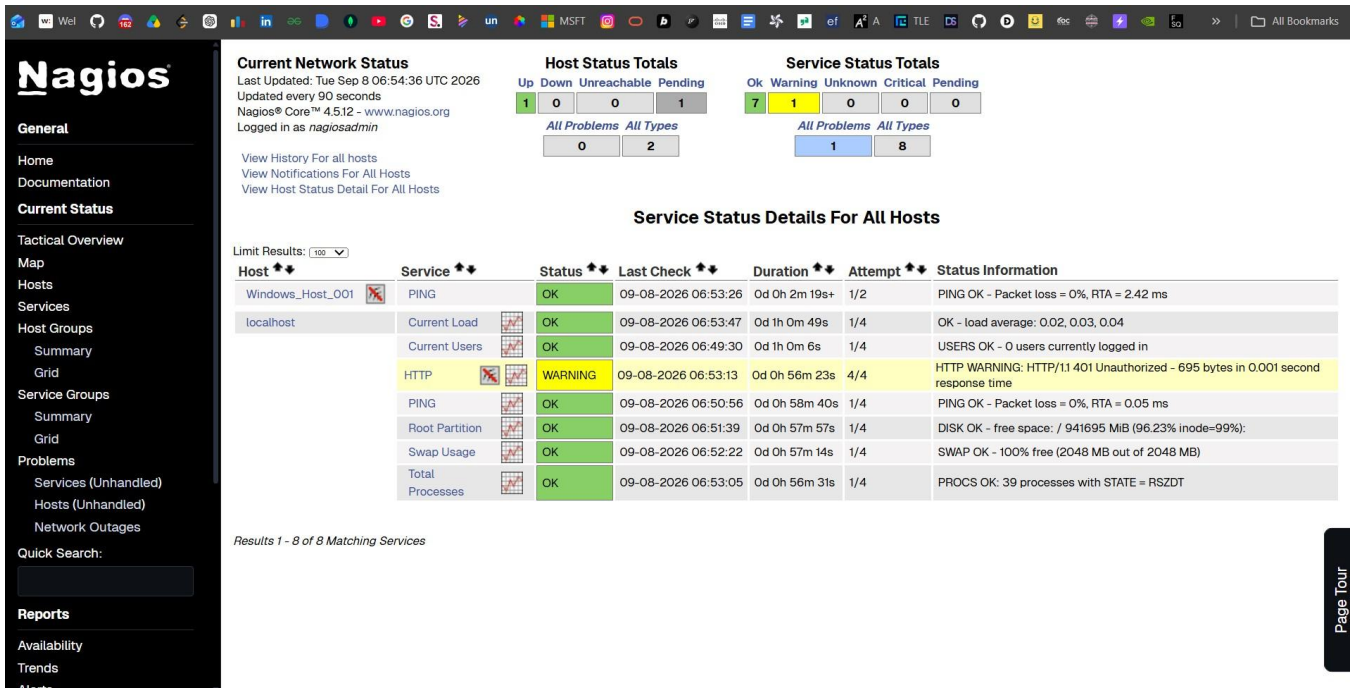
Verify the configuration file for errors, then restart the Nagios container so the changes take effect:

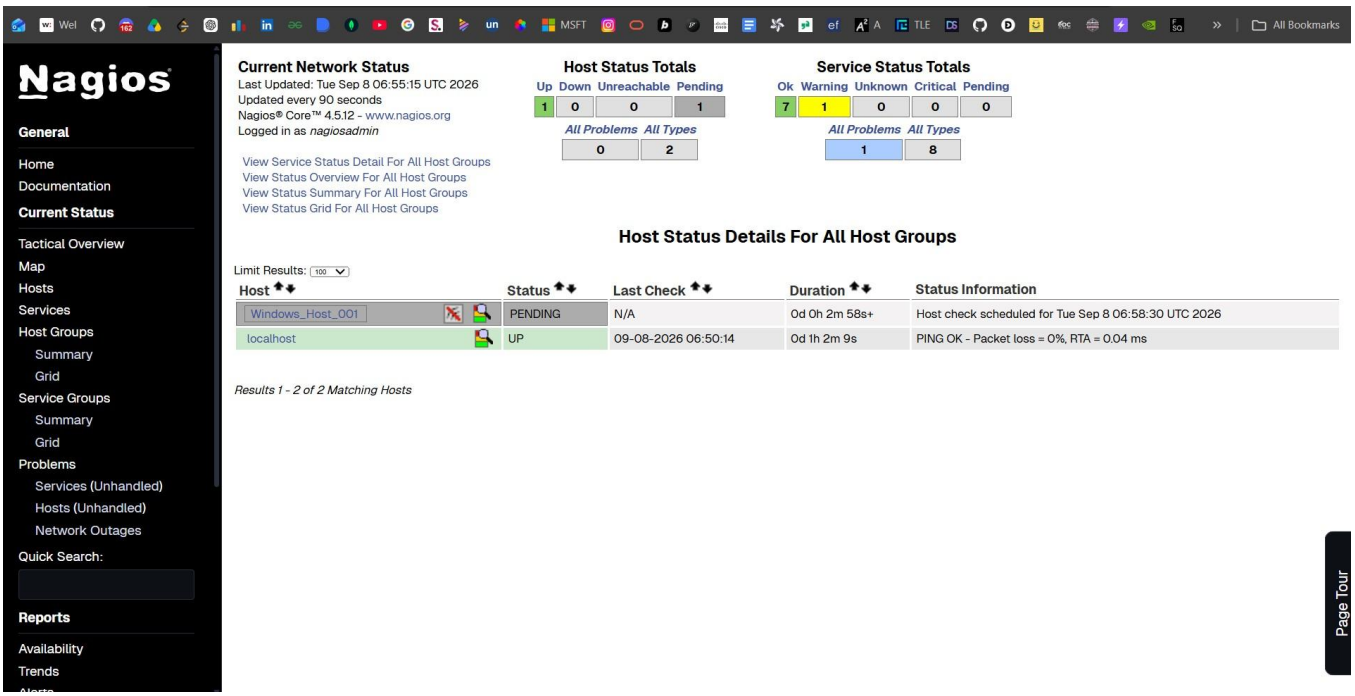


Name: Afifa Syed
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9 - Check Host in Nagios Web Interface

Open the Nagios web interface (http://localhost:8080/nagios/) and check that the new Windows host has been added and is being monitored under Current Status -> Hosts.





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